

Multifractal Analysis for a class of skew product transformations

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- 2 Target systems: Anosov and topological mixing on fibers skew product
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Multifractal analysis

Given a TDS (M, f) and a continuous observable $\varphi : M \rightarrow \mathbb{R}$, then the space M has a natural **multifractal decomposition**

$$M = \bigcup_{\alpha \in \mathbb{R}} M_{\varphi, \alpha} \cup I_{\varphi},$$

where

$$M_{\varphi, \alpha} = \{x \in M : \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{i=0}^{n-1} \varphi(f^i(x)) = \alpha\},$$

and

$$I_{\varphi} = \{x \in M : \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{i=0}^{n-1} \varphi(f^i(x)) \text{ DNE}\}$$

Multifractal analysis

The **Birkhoff average** can be replaced by:

- **Pointwise dimension:** $d_\mu(x) = \lim_{r \rightarrow 0} \frac{\log \mu(B(x, r))}{\log r};$

- **Local entropy:**

$$h_\mu(f, x) = \lim_{\epsilon \rightarrow 0} \lim_{n \rightarrow \infty} -\frac{1}{n} \log \mu(B_n(x, \epsilon));$$

- **Lyapunov exponents:** $\lambda(x) = \lim_{n \rightarrow \infty} \frac{1}{n} \log \|d_x f^n\|;$

and etc.

Multifractal analysis

In ergodic theory, zero measure set is **negligible**. But they may contain **huge and complex dynamics** both from the **topological perspective and dimensional perspective**.

The multifractal analysis aims to investigate the **dimension complexity** of these sets, where the dimension characteristics include **Hausdorff dimension, Bowen's topological entropy (Bowen 1973), Topological pressure (Pesin and Pitskel 1984), packing dimension (Tricot 1982), packing entropy (Huang and Feng 2012)...**

Why call it “multifractal decomposition”?

These decompositions may contain a Cantor-like set:

$$F = \bigcap_{n=0}^{\infty} \bigcup_{(i_0 \dots i_n)} \Delta_{i_0 \dots i_n}$$

where the basic sets on the n -th step $\Delta_{i_0 \dots i_n}$ are closed and satisfying

- $\Delta_{i_0 \dots i_n j} \subset \Delta_{i_0 \dots i_n}$;
- $\Delta_{i_0 \dots i_n} \cap \Delta_{j_0 \dots j_n} \cap F = \emptyset$ for $(i_0 \dots i_n) \neq (j_0 \dots j_n)$;
- $\text{diam} \Delta_{i_0 \dots i_n} \rightarrow 0$ as $n \rightarrow \infty$.

Cantor-3d

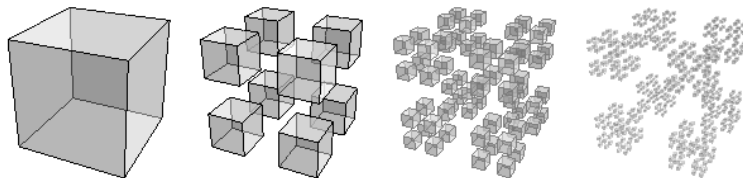


Figure: Cantor dust set

Hausdorff dimension=?

Carathéodory dimension structure

A Carathéodory dimension structure on a set X is given by the following data:

- ① An indexed collection of subsets of X , denote $\mathcal{F} = \{U_s : s \in \mathcal{S}\}$,
- ② Functions $\xi, \eta, \psi : \mathcal{S} \rightarrow \infty[0, \infty)$ satisfying the following conditions:
 - If $U_s = \emptyset$, then $\eta(s) = \psi(s) = 0$; if $U_s \neq \emptyset$, then $\eta(s), \psi(s) > 0$;
 - for any $\epsilon > 0$, one can find $\delta > 0$ such that $\eta(s) \leq \epsilon$ whenever $\psi(s) \leq \delta$;
 - for any $\epsilon > 0$, there exists a finite or countable subcollection $\mathcal{G} \subset \mathcal{S}$ such that $X \subset \cup_{s \in \mathcal{G}} U_s$ and $\sup_{s \in \mathcal{G}} \psi(s) \leq \epsilon$.

Here, $\xi(s)$ can be thought as the **weight** of U_s , and $\eta(s)$ and $\psi(s)$ can be interpreted as a **size or scale** of U_s .

Carathéodory dimension structure

For a nonempty set $Z \subset X$ and $\alpha \in \mathbb{R}$, set

$$m_C(Z, \alpha) := \lim_{\epsilon \rightarrow 0} \inf_{\mathcal{G}} \sum_{s \in \mathcal{G}} \xi(s) \eta(s)^\alpha,$$

where the infimum is taken over all finite or countable $\mathcal{G} \subset \mathcal{S}$ covering Z with $\sup_{s \in \mathcal{G}} \psi(s) \leq \epsilon$.

Proposition

For any nonempty set $Z \subset X$, there exists a critical value $\alpha_C \in \mathbb{R}$ such that $m_C(Z, \alpha) = \infty$ for $\alpha < \alpha_C$ and $m_C(Z, \alpha) = 0$ for $\alpha > \alpha_C$.

Here, the $\alpha_C = \dim_C(Z)$ is called **the Carathéodory dimension** of Z associated to the C-structure $(\mathcal{S}, \mathcal{F}, \xi, \eta, \psi)$.

Examples of C-structure: Hausdorff dimension

Hausdorff dimension: $\mathcal{S} = X \times (0, \infty)$, $U_s = B(x, r)$ for $s = (x, r) \in \mathcal{S}$, $\xi(x, r) = 1$, $\eta(x, r) = \psi(x, r) = \text{diam}(B(x, r))$. Let

$$m_C(Z, \alpha) = \liminf_{\epsilon \rightarrow 0} \left\{ \sum_i (\text{diam}(B(x_i, r_i)))^\alpha : Z \subset \cup_i B(x_i, r_i), r_i < \epsilon \right\}$$

The critical value is the Hausdorff dimension.

Examples of C-structure: Bowen's top. entropy & pressure

Topological entropy (Bowen, 1973) and topological pressure (Pesin and Pitskel, 1984):

For $r > 0$, let $\mathcal{S} = X \times \mathbb{N}$, $U_s = B_n(x, r)$ for $s = (x, n)$, $\xi(x, n) = e^{S_n\varphi(x)}$, $\eta(x, n) = e^{-n}$, $\psi(x, n) = \frac{1}{n}$. Then

$$m_C^r(Z, \alpha) = \lim_{N \rightarrow \infty} \inf_{\mathcal{G}, n \geq N} \sum_{(x, n) \in \mathcal{G}} e^{S_n\varphi(x)} e^{-n\alpha}.$$

Writing $\dim_C^r(Z)$ for the critical value of α , then the topological pressure of φ on the set Z is

$$P_Z(\varphi) = \limsup_{r \rightarrow 0} \dim_C^r(Z).$$

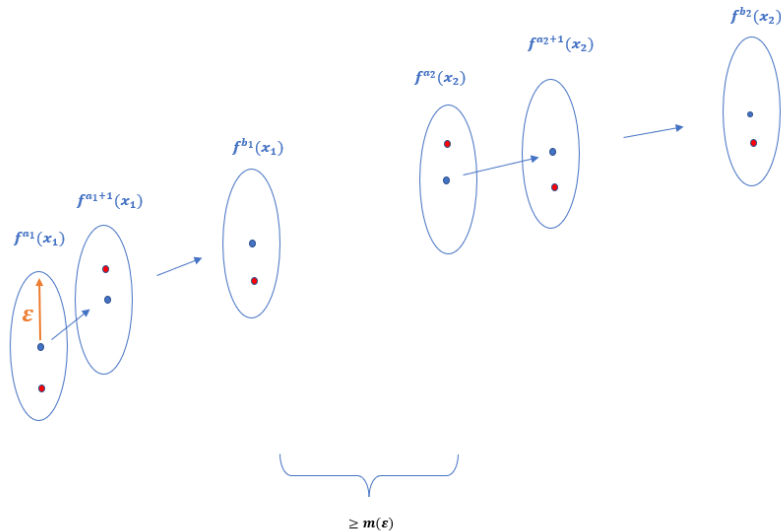
If Z is compact and invariant, then they coincides with the classical definition utilizing separated set or covering set.

Existing results: Multifractal analysis on Birkhoff average

Approaches:

- Thermodynamic approach based on existence and uniqueness of equilibrium states → Climenhaga 2014 Survey;
- orbit-gluing approach → Specification property (or weak form).

Specification property



Existing results: Multifractal analysis on Birkhoff average

A variational principle:

$$h_{top}(M_{\varphi,\alpha}) = \sup\{h_{\mu}(f) : \mu \in \mathcal{M}_f(M) \text{ and } \int \varphi d\mu = \alpha\}$$

has been established for:

- Fan, Feng, Wu 2001: topological mixing SFT;
- Takens and Verbitskiy 2003: maps with the specification property;
- Pfister and Sullivan 2007: maps with g-almost product property;
- Tian and Varandas 2017: only “ $\geq$ ” for maps with non-uniform specification.

A generalised variational principle regarding top. pressure has also been established by Thompson 2009 for maps with specification property; Chen and Pei 2010 for g-almost product property; Chen, Zheng and Zhou 2016 for maps with non-uniform specification.

Existing results: Multifractal analysis on Birkhoff average

For skew product system and Random dynamical system:

- Feng and Shu 2009: product of a ergodic transformation on a Lebesgue space and a full shift of finite type + equicontinuous function $\rightarrow$ Multifractal formalism + variational principle about Hausdorff dimension of conditional level sets.
- Shu 2010: for random perturbation of mixing smooth conformal repeller, the Hausdorff dimension of conditional level sets of Birkhoff average $\rightarrow$ Hausdorff dimension of Birkhoff average for the unperturbed system.

Existing results: Multifractal analysis on Birkhoff average

If I_φ is nonempty, then it carries

- Pesin and Pitskel 1984: full top. entropy for full shift on two symbols;
- Barreira and Schmeling 2000: full top entropy and Hausdorff dimension for top. mixing SFT, conformal repellers and conformal horseshoe (Hölder continuous potential);
- Fan, Feng and Wu 2001: full top. entropy and Hausdorff dimension for top. mixing SFT;
- Chen, Tassilo and Shu 2005: full top. entropy for maps with specification;
- Thompson 2010 & 2012: full top. entropy for maps with specification & almost specification, e.g. β -transformation;
- Dong, Oprocha and Tian 2018: full top. entropy for maps with shadowing property.
- Barrientos, Nakano, Raibekas and Roldan 2022: full top. entropy and Hausdorff dimension for some non-hyperbolic dynamical systems.

Dynamical system driven by an external force

M : connected compact smooth Riemannian manifold without boundary;
 $\text{Diff}^2(M)$: the space of C^2 –diffeomorphisms on M equipped with the C^2 –topology;
 $\theta : \Omega \rightarrow \Omega$: a homeomorphism on a compact metric space Ω ;
Let $f : \Omega \mapsto \text{Diff}^2(M)$ be a continuous map.

Dynamical system driven by an external force

The diffeomorphism $f_\omega := f(\omega)$ is driven by an external force (Ω, θ) , i.e. while ω is shifted by θ in time n to $\theta^n \omega$ on the external force space Ω , any point $x \in M$ is mapping to

$$F(n, \omega)x = f_\omega^n x := \begin{cases} f_{\theta^{n-1}\omega} \circ \cdots \circ f_{\theta\omega} \circ f_\omega x, & \text{if } n > 0 \\ x, & \text{if } n = 0 \\ (f_{\theta^n \omega})^{-1} \circ \cdots \circ (f_{\theta^{-1}\omega})^{-1} x, & \text{if } n < 0. \end{cases}$$

The following system

$$F : \mathbb{Z} \times \Omega \times M \rightarrow M, (n, \omega, x) \mapsto F(n, \omega)x$$

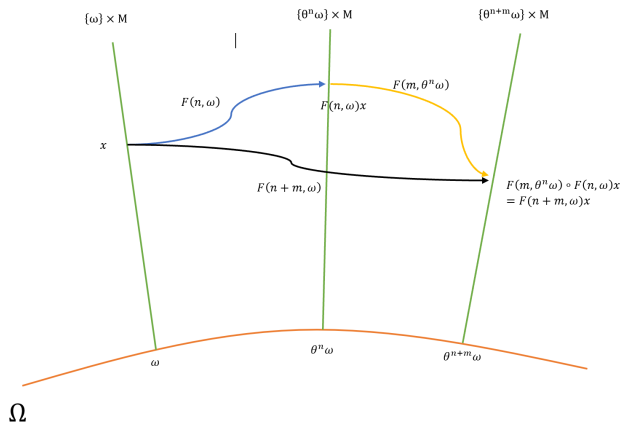
satisfies that for each $n \in \mathbb{Z}$, $(\omega, x) \mapsto F(n, \omega)x$ is continuous and the mappings $F(n, \omega) : M \rightarrow M$ form a cocycle over θ , i.e.,

$$F(0, \omega) = id_M \text{ for all } \omega \in \Omega,$$

$$F(n + m, \omega) = F(m, \theta^n \omega) \circ F(n, \omega) \text{ for all } n, m \in \mathbb{Z}, \omega \in \Omega.$$

When $(\Omega, \mathcal{B}(\Omega))$ is equipped with a θ –invariant probability measure P , the system F is a **continuous random dynamical system**[Arnold].

cocycle property

Figure: cocycle property moving on the bundle $\Omega \times M$

Skew product

Putting f_ω and θ together forms a skew product map $\Theta : M \times \Omega \rightarrow M \times \Omega$

$$\Theta(x, \omega) = (f_\omega(x), \theta\omega) \text{ for all } (x, \omega) \in M \times \Omega.$$

Anosov on Fibers

Θ is called **Anosov on fibers** if for every $(x, \omega) \in M \times \Omega$, there is a splitting of the tangent bundle of $M_\omega := M \times \{\omega\}$ at x

$$T_x M_\omega = E_{(x, \omega)}^s \oplus E_{(x, \omega)}^u$$

which depends continuously on $(x, \omega) \in M \times \Omega$ with $\dim E_{(x, \omega)}^s$, $\dim E_{(x, \omega)}^u > 0$, and the splitting is invariant in the sense that

$$D_x f_\omega E_{(x, \omega)}^u = E_{\phi(x, \omega)}^u, \quad D_x f_\omega E_{(x, \omega)}^s = E_{\phi(x, \omega)}^s$$

and

$$\begin{cases} |D_x f_\omega \xi| \geq e^{\lambda_0} |\xi|, & \forall \xi \in E_{(x, \omega)}^u, \\ |D_x f_\omega \eta| \leq e^{-\lambda_0} |\eta|, & \forall \eta \in E_{(x, \omega)}^s, \end{cases}$$

where $\lambda_0 > 0$ is a constant.

Topological Mixing on Fibers

Θ is said to be **topological mixing on fibers** if for any nonempty open sets $U, V \subset M$, there exists $N > 0$ such that for any $n \geq N$ and $\omega \in \Omega$,

$$\Theta^n(\{\omega\} \times U) \cap (\{\theta^n \omega\} \times V) \neq \emptyset.$$

Example (Fiber Anosov maps on 2 – d tori)

Let $\Omega := \mathbb{T}$ and let $\theta : \Omega \rightarrow \Omega$ be irrational rotation $\theta(\omega) = \omega + \alpha$ for some $\alpha \in \mathbb{R} - \mathbb{Q}$.
Let $M := \mathbb{T}^2$.

Define

$$f_{\omega} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} h_1(\omega) \\ h_2(\omega) \end{pmatrix} \text{ for } \begin{pmatrix} x \\ y \end{pmatrix} \in \mathbb{T}^2$$

where $h_1(\omega)$, $h_2(\omega)$ are continuous map from $\mathbb{T}$ to itself.

Example (Random composition of 2×2 area-preserving positive matrices)

Let

$$\left\{ A_i = \begin{pmatrix} a_i & b_i \\ c_i & d_i \end{pmatrix} \right\}_{1 \leq i \leq k}$$

be 2×2 matrices with positive integer entries and $|\det A_i| = 1$ for all $i \in \{1, \dots, k\}$. Let $\Omega := S_k = \{1, \dots, k\}^{\mathbb{Z}}$ together with the left shift operator θ be the symbolic dynamical system with k symbols.

Define $f : \Omega \rightarrow \{A_1, \dots, A_k\}$ by $f_\omega = A_{\omega(0)} : \mathbb{T}^2 \rightarrow \mathbb{T}^2$ for $\omega = (\dots, \omega(-1), \omega(0), \omega(1), \dots) \in S_k$.

$$f_\omega^n = \begin{cases} A_{\omega(n-1)} \circ \dots \circ A_{\omega(1)} \circ A_{\omega(0)}, & \text{if } n > 0 \\ id, & \text{if } n = 0 \\ A_{\omega(n)}^{-1} \circ \dots \circ A_{\omega(-2)}^{-1} \circ A_{\omega(-1)}^{-1}, & \text{if } n < 0. \end{cases}$$

Dynamical complexity from the geometric and topological perspective

Theorem (Wen Huang, Kening Lu, Zeng Lian, ArXiv 2019)

Assume that Θ is Anosov and topological mixing on fibers driven by a homeomorphism. Then,

- ① *Random periodic points are dense in $L^\infty(\Omega, M)$;*
- ② *Θ^N has k -symbol strong full random horseshoe for some N ;*
- ③ *Periodic invariant measure is dense in invariant measures with respect to the narrow topology;*
- ④ *Random specification property.*

Dynamical complexity from the statistical perspective

Theorem (L., ArXiv 2021)

There exists a constant $\nu_0 \in (0, 1)$ only depending on the system Θ such that for any $\gamma, \nu \in (0, 1)$ satisfying

$$0 < \gamma + \nu < \nu_0$$

and $\psi \in C^{0,\gamma}(M)$, $\varphi \in C^{0,\nu}(M)$, both the (quenched) past and future random correlation of φ and ψ exponential decay with respect to Θ and the unique random SRB measure μ , i.e. for any $n \in \mathbb{N}$, $\omega \in \Omega$,

$$\begin{aligned} \left| \int_M \psi(f_{\theta^{-n}\omega}^n x) \varphi(x) d\mu_{\theta^{-n}\omega} - \int_M \psi d\mu_\omega \int_M \varphi d\mu_{\theta^{-n}\omega} \right| &\leq K \|\psi\|_{C^{0,\gamma}(M)} \cdot \|\varphi\|_{C^{0,\nu}(M)} \cdot \Lambda^n; \\ \left| \int_M \psi(f_\omega^n x) \varphi(x) d\mu_\omega - \int_M \psi d\mu_{\theta^n\omega} \int_M \varphi d\mu_\omega \right| &\leq K \|\psi\|_{C^{0,\gamma}(M)} \cdot \|\varphi\|_{C^{0,\nu}(M)} \cdot \Lambda^n, \end{aligned}$$

where positive constants K and $\Lambda \in (0, 1)$ only depends on γ and ν .

How about the dynamical complexity from the **dimensional perspective?**

Fiber Bowen's topological entropy

For $n \in \mathbb{N}$, $\omega \in \Omega$, we denote the fiber Bowen's metric

$$d_{\omega}^n(x, y) = \max_{0 \leq i \leq n-1} \{d_M(f_{\omega}^i x, f_{\omega}^i y)\} \text{ for } x, y \in M.$$

Let $B_n(\omega, x, \epsilon) \subset M$ be the open ball of radius ϵ around $x \in M$ with respect to the metric d_{ω}^n . For any $\epsilon > 0$, $s \in \mathbb{R}$, on the fiber $\{\omega\} \times M$, for any nonempty set $Z \subset M$, we define

$$m(Z, s, \omega, N, \epsilon) = \inf_{\Gamma_{\omega}^{\epsilon}} \sum_i e^{-sn_i},$$

where the infimum is taken over all finite or countable collections $\Gamma_{\omega}^{\epsilon} = \{B_{n_i}(\omega, x_i, \epsilon)\}$ with $Z \subset \cup_i B_{n_i}(\omega, x_i, \epsilon)$ and $\min\{n_i\} \geq N$. Note that $m(Z, s, \omega, N, \epsilon)$ does not decrease as N increases, the following limit exists

$$m(Z, s, \omega, \epsilon) = \lim_{N \rightarrow \infty} m(Z, s, \omega, N, \epsilon).$$

Fiber Bowen's topological entropy: Special case of C-structure

There exists a number $h_{top}(F, Z, \omega, \epsilon) \in [0, +\infty)$ such that

$$m(Z, s, \omega, \epsilon) = \begin{cases} +\infty, & \text{if } s < h_{top}(F, Z, \omega, \epsilon) \\ 0, & \text{if } s > h_{top}(F, Z, \omega, \epsilon). \end{cases}$$

Note that $m(Z, h_{top}(F, Z, \omega, \epsilon), \omega, \epsilon)$ could be $+\infty$, 0 or positive finite number.

Lemma

The value $h_{top}(F, Z, \omega, \epsilon)$ does not decrease as ϵ decreases. Therefore, the following limit exists

$$h_{top}(F, Z, \omega) = \lim_{\epsilon \rightarrow 0} h_{top}(F, Z, \omega, \epsilon) = \sup_{\epsilon > 0} h_{top}(F, Z, \omega, \epsilon).$$

Conditional Level sets of Birkhoff average

For any continuous observable $\varphi : \Omega \times M \rightarrow \mathbb{R}$ and $\alpha \in \mathbb{R}$, denote

$$K_{\varphi, \alpha} = \{(\omega, x) \in \Omega \times M : \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{i=0}^{n-1} \varphi(\Theta^i(\omega, x)) = \alpha\}$$

and conditional level set of Birkhoff average

$$K_{\varphi, \alpha}(\omega) = \{x \in M : (\omega, x) \in K_{\varphi, \alpha}\}.$$

Denote $\Omega_\alpha = \{\omega \in \Omega : K_{\varphi, \alpha}(\omega) \neq \emptyset\}$. If (Ω, θ) is **uniquely ergodic with unique invariant measure $\mathbb{P}$ and complete Borel σ -algebra $\mathcal{B}_{\mathbb{P}}(\Omega)$** , then Ω_α is a measurable set and either $\mathbb{P}(\Omega_\alpha) = 1$ or $\mathbb{P}(\Omega_\alpha) = 0$.
Denote $L_\varphi := \{\alpha \in \mathbb{R} : \mathbb{P}(\Omega_\alpha) = 1\}$.

Variational principle on Birkhoff average

Theorem (Nian Liu, and L., preprint)

Assume $(\Omega \times M, \Theta)$ is Anosov and topological mixing on fibers system driven by a uniquely ergodic homeomorphism system $(\Omega, \mathcal{B}_{\mathbb{P}}(\Omega), \mathbb{P}, \theta)$. Let $\varphi \in C(\Omega \times M, \mathbb{R})$, for $\alpha \in L_{\varphi}$, then the following holds for $\mathbb{P}$ -a.s. $\omega \in \Omega$:

$$h_{\text{top}}(F, K_{\varphi, \alpha}(\omega), \omega) = \max\{h_{\mu}(F) : \mu \in I_{\Theta}(\Omega \times M, \varphi, \alpha)\}.$$

Here $h_{\mu}(F)$ is the fiber measure theoretic entropy of F (or relative measure theoretic entropy of Θ). The Abramov-Rohlin formula:

$$h_{\mu}(\Theta) = h_{\mu}(F) + h_{\mathbb{P}}(\theta), \quad \mu \in I_{\Theta}(\Omega \times M).$$

Irregular sets

For any $\varphi \in C(\Omega \times M, \mathbb{R})$, the irregular set of φ is defined by

$$I_\varphi = \left\{ (\omega, x) \in \Omega \times M : \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{i=0}^{n-1} \varphi(\Theta^i(\omega, x)) \text{ does not exist} \right\},$$

and the fiber of I_φ is defined by $I_\varphi(\omega) = \{x \in M : (\omega, x) \in I_\varphi\}$.

Irregular sets

Theorem (Nian Liu and L., preprint)

Assume $(\Omega \times M, \Theta)$ is Anosov and topological mixing on fibers system driven by a uniquely ergodic homeomorphism system $(\Omega, \mathcal{B}_{\mathbb{P}}(\Omega), \mathbb{P}, \theta)$. Then for any $\varphi \in C(\Omega \times M, \mathbb{R})$, if $I_{\varphi}(\omega^) \neq \emptyset$ for some $\omega^* \in \Omega$, then*

- 1 *the fiber of irregular set is nonempty for $\mathbb{P}$ -a.e. $\omega \in \Omega$, that is:*

$$\mathbb{P}(\{\omega \in \Omega : I_{\varphi}(\omega) \neq \emptyset\}) = 1.$$

- 2 *$\mathbb{P}$ -a.e. fiber of the irregular set carries full fiber topological entropy:*

$$h_{\text{top}}(F, I_{\varphi}(\omega), \omega) = h_{\text{top}}(F) \text{ for } \mathbb{P} - \text{a.e. } \omega \in \Omega.$$

Fiber specification property

Definition

For any $\omega \in \Omega$, an ω -specification $S_\omega = (\omega, \tau, P_\omega)$ consists of a finite collection of intervals $\tau = \{I_1, \dots, I_k\}$, $I_i = [a_i, b_i] \subset \mathbb{Z}$, and a map $P_\omega : \cup_{i=1}^k I_i \rightarrow M$ such that for $t_1, t_2 \in I \in \tau$,

$$F_{\theta^{t_1}\omega}^{t_2-t_1}(P_\omega(t_1)) = P_\omega(t_2).$$

An ω -specification S_ω is called m -spaced if $a_{i+1} > b_i + m$ for all $i \in \{1, \dots, k-1\}$.

The system Θ is said to have the fiber specification property if for any $\epsilon > 0$, there exists $m = m(\epsilon) > 0$ such that for any $\omega \in \Omega$, any m -spaced ω -specification $S_\omega = (\omega, \tau, P_\omega)$, we can find a point $x \in M$, which is (ω, ϵ) -shadowing S_ω , i.e.,

$$d_M(P_\omega(t), F_\omega^t x) < \epsilon \text{ for any } t \in I \in \tau.$$

Fiber specification property

Theorem (Nian Liu, and L., preprint)

Assume that $(\Omega \times M, \Theta)$ is Anosov on fibers and topological mixing on fibers system driven by a homeomorphism (Ω, θ) , then Θ has the fiber specification property.

Outline of Proof

Lemma (A Key Lemma)

Let $\alpha = \int \varphi d\mu$ for any $\mu \in I_\Theta(\Omega \times M)$. For any $\gamma > 0$ and positive number $\delta < \frac{1}{2}\{\gamma, \eta\}$, there exists a $\mathbb{P}$ -full measure set Ω_δ such that for any $N' \in \mathbb{N}$, there exists a **measurable** function $\hat{n} : \Omega_\delta \rightarrow \mathbb{N}$ satisfying for all $\omega \in \Omega_\delta$, $\hat{n}(\omega) \geq N'$,

$$P(\alpha, 4\delta, \hat{n}(\omega), \omega) := \left\{ x \in M : \left| \frac{1}{\hat{n}(\omega)} \sum_{i=0}^{\hat{n}(\omega)-1} \varphi(\Theta^i(\omega, x)) - \alpha \right| < \delta \right\} \neq \emptyset,$$

and

$$\frac{1}{\hat{n}(\omega)} \log M(\alpha, 4\delta, \hat{n}(\omega), \frac{\eta}{2}, \omega) \geq h_\mu(F) - 4\gamma,$$

where $\eta > 0$ is the fiber-expansive constant and $M(\alpha, 4\delta, \hat{n}(\omega), \frac{\eta}{2}, \omega)$ is the largest cardinality of maximal $(\omega, \frac{\eta}{2}, \hat{n}(\omega))$ -separated sets of $P(\alpha, 4\delta, \hat{n}(\omega), \omega)$.

Outline of Proof

For $k \in \mathbb{N}$, Define $m_k = m(\frac{\eta}{2^{4+k}})$ and pick $\delta_k \searrow 0$. By the key lemma, there exists $\mathbb{P}$ -full measure set Ω_{δ_k} and a measurable $\hat{n}_k : \Omega_{\delta_k} \rightarrow \mathbb{N}$ satisfying

- $\hat{n}_k(\omega) \geq 2^{m_k}$,
- $P(\alpha, 4\delta_k, \hat{n}_k(\omega), \omega) = \left\{ x \in M : \left| \frac{1}{\hat{n}_k(\omega)} \sum_{i=0}^{\hat{n}_k(\omega)-1} \varphi(\Theta^i(\omega, x)) - \alpha \right| < \delta_k \right\} \neq \emptyset$,
- $\frac{1}{\hat{n}_k(\omega)} \log M(\alpha, 4\delta_k, \hat{n}_k(\omega), \frac{\eta}{2}, \omega) \geq h_{\mu_0}(F) - 4\gamma$.

Question: How to gluing?

Caution: $\hat{n}_k(\omega)$ may be very large!

Outline of Proof

Pick $\{\xi_k\}_{k=1}^\infty \searrow 0$ with $\xi_1 < \frac{1}{2}$. Now for any $k \in \mathbb{N}$, by using Lusin's theorem, there exists a compact set $\Omega_k \subset \cap_{j \geq 1} \Omega_{\delta_j}$ with $\mathbb{P}(\Omega_k) \geq 1 - \frac{\xi_k}{2}$, such that

$\hat{n}_k(\omega)$ is **continuous** on Ω_k , hence **bounded**!

By Birkhoff's Ergodic theorem, there exists a $\mathbb{P}$ -full measure set Ω'_k , such that for any $\omega \in \Omega'_k$, we have

$$\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{i=0}^{n-1} 1_{\Omega_k}(\theta^i \omega) = P(\Omega_k) \geq 1 - \frac{\xi_k}{2}.$$

We define $\tilde{\Omega}_\gamma := \cap_{k \geq 1} \Omega'_k$,

Outline of proof

Constructing Moran-like fractal:

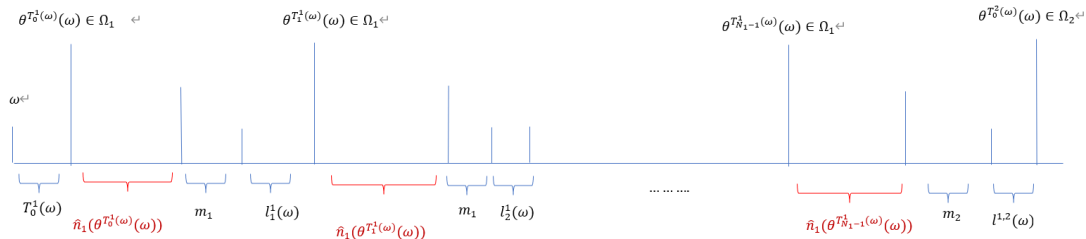


Figure: Constructing the centers of balls forming the first level set of fiber Moran-like fractal

Outline of proof

- Centers of balls forming the k -th level of fiber Moran-like fractal: $H_k(\omega)$;
- let $\mathfrak{X}_k(\omega) = \bigcup_{x \in H_k(\omega)} \overline{B}_{T_{N_k(\omega)}^k(\omega)}\left(\omega, x, \frac{\eta}{2^{4+k}}\right)$;
- the fiber Moran-like fractal: $\mathfrak{X}(\omega) = \bigcap_{k \geq 1} \mathfrak{X}_k(\omega) \subset K_{\varphi, \alpha}(\omega)$;
- construct a measure: $\mu_{k, \omega} = \frac{1}{\#H_k(\omega)} \sum_{x \in H_k(\omega)} \delta_x \rightarrow \mu_\omega$ in weak* topology;
- entropy distribution principle.

Multifractal formalism on Birkhoff average

Theorem (Nian Liu and L., preprint)

Assume $(\Omega \times M, \Theta)$ is Anosov and topological mixing on fibers system driven by a uniquely ergodic homeomorphism system $(\Omega, \mathcal{B}_{\mathbb{P}}(\Omega), \mathbb{P}, \theta)$. Let $\varphi \in C(\Omega \times M, \mathbb{R})$, then

(a) for any $\alpha \in L_{\varphi}$, one has

$$h_{\text{top}}(F, K_{\varphi, \alpha}(\omega), \omega) \leq \inf_{q \in \mathbb{R}} \{ \pi_F(q\varphi) - q\alpha \} \text{ for } \mathbb{P} - \text{a.s. } \omega \in \Omega. \quad (1)$$

(b) for $\alpha \in \text{int}(L_{\varphi})$, the above equality holds.

Here $\pi_F(\varphi)$ denotes the fiber topological pressure

$$\pi_F(\varphi) = \lim_{\epsilon \rightarrow 0} \limsup_{n \rightarrow \infty} \frac{1}{n} \int \log \left(\sup_Q \sum_{x \in Q} \exp \left(\sum_{i=0}^{n-1} \varphi(\Theta^i(\omega, x)) \right) \right) d\mathbb{P}(\omega).$$

Specification for RDS

The following specification was defined by Gundlach and Kifer 2000.

Let $\epsilon_0 : \Omega \rightarrow \mathbb{R}^+$ be the expansive characteristic for RDS F , i.e., for any $\omega \in \Omega$

if $d_M(F_\omega^n(x), F_\omega^n(y)) \leq \epsilon_0(\theta^n \omega)$ for $n \in \mathbb{Z}$, then $x = y$.

The RDS F has k -specification if for each constant $c > 0$ and $\mathbb{P}$ -a.s. $\omega \in \Omega$, there exists an $\mathbb{N}$ -valued random variable $L_c = L_c(\omega) \geq 1$ such that for any points $\{x_i\}_{i=0}^k \in M$ and integers $-\infty \leq b_0 < a_1 < b_1 < \dots < a_k < b_k < a_{k+1} \leq \infty$ satisfying $a_{i+1} \geq b_i + L_c(\theta^{b_i} \omega)$, $i = 0, 1, \dots, k$ one can find $z \in M$ such that

$$\max_{a_i \leq j \leq b_i} d_M(F_\omega^j(x_i), F_\omega^j(z)) \leq c\epsilon_0(\theta^j \omega) \quad \forall i = 1, \dots, k \quad (2)$$

$$d_M(F_\omega^j(x_0), F_\omega^j(z)) \leq c\epsilon_0(\theta^j \omega) \quad \forall j \geq a_{k+1} \text{ and } \forall j \leq b_0. \quad (3)$$

If F satisfies k -specification for any $k \in \mathbb{N}_+$ with L_c independent of k , then we say that F has specification property.

Multifractal analysis on Local entropy

Theorem (Nian Liu and L., preprint)

Assume that $(\Omega \times M, \Theta)$ is Anosov on fibers and topological mixing on fibers system driven by a homeomorphism (Ω, θ) , then Θ has the specification property given by Gundlach and Kifer.

As a corollary, for any Hölder continuous potential $\varphi : \Omega \times M \rightarrow \mathbb{R}$, there exists a unique equilibrium state, named $\mu = \mu_\varphi \in I_\Theta^e(\Omega \times M)$, satisfying the Gibbs property, i.e., for any $\epsilon \in (0, \frac{\eta}{4})$, there exist positive constants A_ϵ, B_ϵ such that for P -a.s. $\omega \in \Omega$,

$$A_\epsilon \leq \mu_\omega(B_n(\omega, x, \epsilon)) \cdot \pi_F(\varphi)(\omega, \epsilon, n) \cdot e^{-\sum_{i=0}^{n-1} \varphi(\Theta^i(\omega, x))} \leq B_\epsilon \text{ for all } x \in M, \quad (4)$$

where $\eta > 0$ is a fiber expansive constant.

Multifractal analysis on Local entropy

Corollary (Nian Liu and L., preprint)

Assume $(\Omega \times M, \Theta)$ is Anosov and topological mixing on fibers system driven by a uniquely ergodic homeomorphism system $(\Omega, \mathcal{B}_P(\Omega), \mathbb{P}, \theta)$. Let φ be a Hölder potential and $\mu = \mu_\varphi$ be the Gibbs state. For $\alpha \in \mathbb{R}$, we denote

$$E_\alpha = \{(\omega, x) \in \Omega \times M : \bar{h}_\mu(F; \omega, x) = \underline{h}_\mu(F; \omega, x) = \pi_F(\varphi) - \alpha\}.$$

Then

① for μ -a.s. $(\omega, x) \in \Omega \times M$,

$$\bar{h}_\mu(F; \omega, x) = \underline{h}_\mu(F; \omega, x) = h_\mu(F) = \pi_F(\varphi) - \int \varphi d\mu;$$

② for $\alpha \in L_\varphi$,

$$h_{\text{top}}(F, E_\alpha(\omega), \omega) = \max\{h_\mu(F) : \mu \in I_\Theta(\Omega \times M, \varphi, \alpha)\} \text{ for } \mathbb{P} - \text{a.s. } \omega \in \Omega;$$

③ for any $\alpha \in L_\varphi$,

$$h_{\text{top}}(F, E_\alpha(\omega), \omega) \leq \inf_{q \in \mathbb{R}} \{\pi_F(q\varphi) - q\alpha\} \text{ for } \mathbb{P} - \text{a.s. } \omega \in \Omega;$$

④ for $\alpha \in \text{int}(L_\varphi)$,

$$h_{\text{top}}(F, E_\alpha(\omega), \omega) = \inf_{q \in \mathbb{R}} \{\pi_F(q\varphi) - q\alpha\} \text{ for } \mathbb{P} - \text{a.s. } \omega \in \Omega.$$

Thanks For Your Attention!